

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

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Course Code & Title:	21CSC201J - DATA STRUCTURES AND ALGORITHMS (BIG DATA)
Task & Channel:	Assignment on Hashing Techniques & Collision Resolution [TEAMS]

Assignment Description:

Explain Double Hashing vs Quadratic Probing. Simulate insertion of keys [12, 44, 13, 88, 23, 94] into table size 11 with $h_1(k) = k \% 11$ and $h_2(k) = 7 - (k \% 7)$.

Technical Solution:

Solution: Double Hashing Collision Resolution

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Hash Functions:

- Primary Hash: $h_1(k) = k \% 11$
- Secondary Hash: $h_2(k) = 7 - (k \% 7)$
- Probe Sequence: $h(k, i) = (h_1(k) + i \cdot h_2(k)) \% 11$

Step-by-Step Table Insertion (Table Size $m = 11$):

1. **Key 12:** $h_1(12) = 12 \% 11 = 1$. Slot 1 empty -> Insert at **[1]**.
2. **Key 44:** $h_1(44) = 44 \% 11 = 0$. Slot 0 empty -> Insert at **[0]**.
3. **Key 13:** $h_1(13) = 13 \% 11 = 2$. Slot 2 empty -> Insert at **[2]**.
4. **Key 88:** $h_1(88) = 88 \% 11 = 0$ (Collision with 44!).
 - $h_2(88) = 7 - (88 \% 7) = 7 - 4 = 3$.
 - $i=1$ implies $(0 + 1 \cdot 3) \% 11 = 3$. Slot 3 empty -> Insert at **[3]**.
5. **Key 23:** $h_1(23) = 23 \% 11 = 1$ (Collision with 12!).
 - $h_2(23) = 7 - (23 \% 7) = 7 - 2 = 5$.
 - $i=1$ implies $(1 + 1 \cdot 5) \% 11 = 6$. Slot 6 empty -> Insert at **[6]**.
6. **Key 94:** $h_1(94) = 94 \% 11 = 6$ (Collision with 23!).
 - $h_2(94) = 7 - (94 \% 7) = 7 - 3 = 4$.
 - $i=1$ implies (6 ...